Soo-Ho Park

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Education

University of British Columbia Sep. 2024 – May 2028
Bachelor of Science: Computer Science *GPA: 94.3%*

- $1500 TREK Excellence Scholarship
- **Related CPSC Courses:** 310 Software Engineering, 313 Computer Hardware and Operating Systems, 304 Relational Databases, 317 Computer Networking, 322 Artificial Intelligence

Langara College Sep. 2022 – Apr. 2024
Associate of Science: Computer Science *GPA: 4.33*

- $2125 Chartwells Awards of Excellence

Experience

Undergraduate Research Assistant | *C++, TLA+* May 2026 – Present
UBC Systopia Lab *Vancouver, BC*

- Built a linearizability checker in C++ that validates execution traces of concurrent data structures by timestamping linearization points of atomic load, store, and CAS operations and totally ordering high-level operations
- Verified the linearizability of a Treiber Stack by sorting operation timestamps, replacing traditional state space searching methodologies that consider up to $n!$ possible orderings of n overlapping operations
- Wrote a formal proof that timestamp ordering always yields a valid linearization, contributing to a research paper
- Presented a research poster at a UBC undergraduate research showcase and refined optimizations through weekly meetings with 2 professors over 5 months

Teaching Assistant | *CPSC 213 – Introduction to Computer Systems* Sep. 2025 – Present
UBC Computer Science *Vancouver, BC*

- Led weekly 1-hour lectures for 30 students on RISC assembly and asynchronous programming in C
- Resolved questions from 100+ students with an average rating of 4.5/5 by running office hours with fellow TAs

Academic Group Tutor | *CPSC 110, MATH 100, 101, 200* Jan. 2025 – Apr. 2025
UBC Alma Mater Society *Vancouver, BC*

- Tutored up to 5 UBC students at a time in computer science and calculus, and arranged 20+ individual appointments for students needing extra help, earning 5/5 ratings from every student in those appointments

Projects

SkyOS | *C++, x86 Assembly, GRUB, QEMU* July 2026 – Present

- Built a freestanding x86 kernel in C++, packaged into a bootable disk image and booted via GRUB (Multiboot specification) under QEMU emulation
- Configured MSVC to produce a non-hosted PE executable compatible with Multiboot's kludge mode by disabling the default CRT and manually adjusting memory layout/alignment to match GRUB's expectations

Cree Learner | *Python, Flask, Azure, TypeScript, React Native* | *SFU StormHacks 2025* Oct. 2025

- Developed a React Native app that helps users learn Plains Cree vocabulary through word memorization with audio pronunciation, built in a 24-hour hackathon
- Built the Flask REST API behind the app, serving word entries from Azure SQL and storing pronunciation audio as blobs in Azure Blob Storage

Vancouver Recreational Activities Scheduler | *Python, Selenium, Flask, Azure SQL, JavaScript* Aug. 2025

- Collaborated in a team of 3 to collect 6,000+ City of Vancouver recreation listings into an Azure SQL database by automating infinite-scroll loading with Selenium and normalizing dates, times, and age ranges with regex
- Enabled users to filter activities by age, weekday, and location by building a Flask REST API with parameterized SQL queries using pyodbc

Technical Skills

Languages: C, C++, Java, Python, SQL, JavaScript, TypeScript, HTML/CSS, R
Frameworks: React Native, Flask, Node.js, JUnit
Developer Tools: Git, Docker, Microsoft Azure (SQL, Blob Storage), QEMU, GRUB, Visual Studio, VS Code
Libraries: Selenium, pyodbc, NumPy